Ashutosh Sathe

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Education

- **Indian Institute of Technology, Bombay** — Powai, India
  Master of Science by Research in Computer Science; GPA: 9.86/10.00 — *Aug 2021 – Jun 2024*
  Recipient of "Institute Silver Medal" and "George B. Fernandes Research Excellence Award" — thesis

Publications

(* = equal contribution)

- Improving Self Consistency in LLMs using Probabilistic Tokenization — arxiv
  Ashutosh Sathe*, Divyanshu Aggarwal*, Sunayana Sitaram — **ICML'24 (W) LLMs and Cognition**
- Efficient Training of Language Models with Compact and Consistent Next Token Distributions — paper
  Ashutosh Sathe, Sunita Sarawagi — **ACL'24, ICML'24 (W) ES-FoMo II**
- MAPLE: Multilingual Evaluation of Parameter Efficient Finetuning of Large Language Models — paper
  Divyanshu Aggarwal*, **Ashutosh Sathe***, Ishaan Watts, Sunayana Sitaram — **ACL'24**
- MEGAVERSE: Benchmarking Large Language Models Across Languages, Modalities, Models and Tasks — paper
  Sanchit Ahuja, Divyanshu Aggarwal, Varun Gumma, Ishaan Watts, **Ashutosh Sathe** *et al.* — **NAACL'24**
- MAFIA: Multi-Adapter Fused Inclusive LanguAge Models — paper
  Prachi Jain*, **Ashutosh Sathe***, Varun Gumma, Kabir Ahuja, Sunayana Sitaram — **EACL'24 (Oral)**
- Benchmarking and Improving Text-to-SQL Generation under Ambiguity — paper, code
  Adithya Bhaskar*, Tushar Tomar*, **Ashutosh Sathe**, Sunita Sarawagi — **EMNLP'23**
- Diverse Parallel Data Synthesis for Cross-Database Adaptation of Text-to-SQL Parsers — paper, code
  Abhijeet Awasthi, **Ashutosh Sathe**, Sunita Sarawagi — **EMNLP'22**

Research Experience

- **Microsoft Research** — Bengaluru, India
  Research Intern — Host: Dr. Sunayana Sitaram — *Sep 2023 - Jun 2024*
  Studied and proposed ways ("MEGAVERSE" and "MAPLE") to improve multilingual capabilities of LLMs. Proposed a technique ("MAFIA") that uses lightweight, modular, reusable components to reduce various societal biases in existing LLMs. Lead a study on probabilistic tokenization which helps improve self consistency in LLM outputs. Also *red-teamed* "Shiksha Copilot" – an AI assistant helping over 10k+ high school teachers in Karnataka.
- **Indian Institute of Science** — Bengaluru, India
  Research Intern — Host: Prof. R. Venkatesh Babu — *May 2019 - Aug 2019*
  Built a robust preprocessor that projects adversarial examples to a "clean" manifold. This boosted robustness of downstrem models against several types of adversarial attacks even in white-box setting.

Projects

- **TestChamber** — code, report, presentation
  Advisor: Prof. Shivaram Kalyanakrishnan — *Course Project: CS748 – Advances in Intelligent and Learning Agents*
  A Gym environment for training RL agents in the popular physics-based puzzle game Portal 2.
- **scarpet-nn** — code, docs, demo
  Efficient Neural Nets in Minecraft — *Personal Project: **125+ stars on GitHub***
  A set of tools to run binarized neural nets in Minecraft. Enables the server admin to load a PyTorch model on any server.
- **yacv** — code, docs, demo
  Yet Another Compiler Visualizer — *Personal Project: **125+ stars on GitHub***
  A tool for visualizing step-by-step parsing process of typical LL(1) and LR parsers using GraphViz and manim.

Skills

Languages Python, C++, C, Go, scarpet
Frameworks PyTorch, HuggingFace Transformers, PyTorch-Geometric, GTK3, OpenGL, JAX (beginner)
Tools Bash, Git, GraphViz, manim, Blender, LaTeX